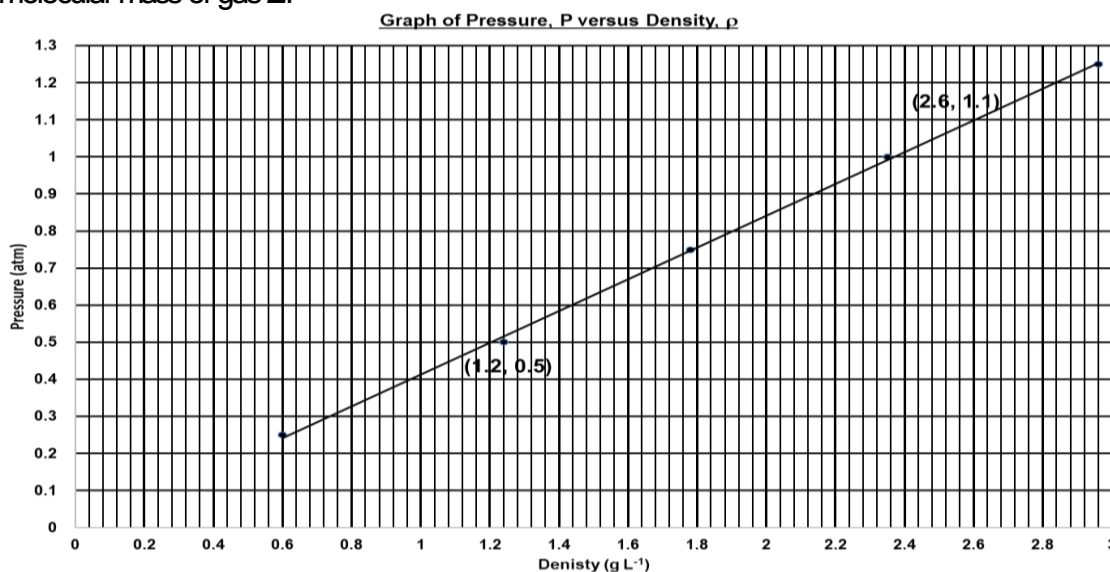


HOUR : 1

1. State the basic assumptions of the kinetic molecular theory of gases of an ideal gas. (CLO1, C2)
2.
 - a) Derive the ideal gas equation from Boyle's law, Charles's law and Avogadro's law.
 - b) A sample of gas occupies a volume of 10 L at the pressure of 2.0 atm. What would be the pressure of the gas if it is allowed to expand in a 50 L container at the same temperature? (0.4 atm)
 - c) A sample of gas occupies 100 mL at 25.0 °C. What volume would the gas occupy at 32.0 °C if the pressure remains constant? (102.35 mL) (CLO3, C3)
3. In the offshore base oil drilling operation, methane gas, CH₄, was found at the bottom of the sea at 480 °C and a pressure of 12.8 atm.
 - a) Calculate the volume of the gas needed to be transferred to a container in the refinery mill with a volume of 1.80×10^3 L at a pressure of 1.00 atm and temperature of 22 °C.
 - b) Determine the mass of the gas being transferred. (358.95 L) (1189.70 g) PSPM 2012 / 2013 (CLO3, C3)
4. Gas C with a mass of 8.00 g at 0 °C and 0.870 atm occupies a volume which is equal to a volume of 11.0 g CO₂ at s.t.p. Calculate the density and the molar mass of the gas. What would be the pressure if the volume of the gas is doubled while the temperature is kept constant. (1.43 g L⁻¹, 36.82 g mol⁻¹, 0.44 atm) PSPM 2006 / 2007 (CLO3, C4)
5.
 - a) A sample of gas A weighing 40.5 g occupies a volume of 15.0 L at 25.0 °C and 1.50 atm. Determine the molar mass of gas A. (44.02 g mol⁻¹)
 - b) What is the volume of oxygen gas at STP required for the complete combustion of 10.0 L of C₂H₄? (30.00 L)
 - c) The density of gas X is 2.60 g L⁻¹ at 25.0 °C and 101 kPa. What is the molar mass of gas X? (63.78 g mol⁻¹) (CLO3, C3)
6. A group of matriculation students carried out an experiment at 25.0 °C to determine the relative molecular mass of gas Z.



Based on the graph above, determine the relative molecular mass of gas Z. (57.06) (CLO3, C3)

HOUR : 2

7. a) A gas mixture containing 2.45 g of N_2 and 3.10 g of Ne occupies a volume of 2.50 L. What is the pressure of the gas at $25.0^\circ C$? (2.36 atm)
- b) 4.00 L of nitrogen at a pressure of 400 kN m^{-2} and 1.00 L of argon at a pressure of 200 kN m^{-2} are transferred into a 2.00 L container. Calculate:
- the partial pressure of nitrogen. (800 kN m^{-2})
 - the partial pressure of argon. (100 kN m^{-2})
 - the total pressure of the mixture. (900 kN m^{-2})
- (CLO3, C3)**
8. When an evacuated glass vessel weighing 134 g was filled with an unknown gas Y, the pressure was found to be 99.3 kN m^{-2} at $31.0^\circ C$, while the mass was 137 g. The glass vessel was then filled completely with water and the mass was 1067 g. By using the ideal gas equation, determine the relative molecular mass of gas Y. (81.85)
- (CLO3, C3)**
9. Calcium carbonate, $CaCO_3$ decomposes upon heating according to the following equation
- $$CaCO_3(s) \xrightarrow{\Delta} CaO(s) + CO_2(g)$$
- 0.20 mol of $CaCO_3$ is heated in a 2.0 dm^3 vessel. After heating, it is found that the pressure of the gas is 44.08 kPa at $30^\circ C$, Determine
- the number of moles of gas formed from the decomposition of $CaCO_3$. (0.035 mol)
 - the mass of $CaCO_3$ left after the reaction was completed. (16.52 g)
 - the partial pressure of hydrogen gas in the vessel when 0.10 mol of hydrogen gas is added to the vessel at the same temperature. (1.24 atm)
- (CLO3, C4)**
10. A gas mixture contains 82% (w/w) methane, CH_4 , and 18% (w/w) ethane, C_2H_6 . If a 15.50 g sample of the gas mixture is placed in a 15L container at $20.0^\circ C$, calculate
- the total moles of gases, (0.887 mol)
 - the total pressure (atm) (1.42 atm)
 - the partial pressure of each gas in the container (1.27 atm, 0.15 atm)
- PSPM 2015 / 2016 (CLO3, C3)**
11. A sample of potassium chlorate, $KClO_3$, was decomposed upon heating, producing potassium chloride, KCl , and oxygen gas, O_2 . The volume of gas collected by displacement of water was 0.250 L at $26^\circ C$ and a pressure of 765 mmHg. Given water vapour pressure at $26^\circ C$ is 25.2 mmHg, calculate
- the partial pressure of O_2 (739.8 mmHg)
 - the moles of O_2 collected ($9.92 \times 10^{-3} \text{ mol}$)
 - the amount of $KClO_3$ (in g) decomposed. (0.81 g)
- (CLO3, C3)**
12. A certain mass of nitrogen gas is added to a 405.0 mL vessel containing 0.5200 g oxygen. The gas pressure increases from 745.0 mmHg to 980.0 mmHg. Assuming that the temperature remains constant and the nitrogen gas does not react with oxygen gas, calculate the mass of the nitrogen gas added. (0.14 g)
- PSPM 2005 / 2006 (CLO3, C4)**

13. **FIGURE 1** shows two connecting vessels containing different gases. When the valve between the vessels is opened, the gases are allowed to mix. Ignoring the volume taken by the valve, calculate the partial pressure of each gas and the final pressure of the mixture at 25 °C.

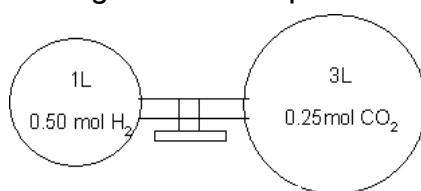


FIGURE 1

(3.06 atm, 1.53 atm, 4.59 atm)

PSPM 2013 / 2014 (CLO3, C3)

HOUR : 3

14. a) State two assumptions made in the ideal gas equation.
b) Explain how assumptions in 14 (a) are addressed for the real gas in the van der Waals

$$\text{equation, } \left(P + \frac{a}{V^2} \right) (V - b) = RT$$

Where a and b are the van der Waals constants.

PSPM 2011 / 2012 (CLO3, C3)

15. An experiment was carried out under high pressure and low temperature to determine the relative molecular mass of CO₂ gas. The following data were obtained.

Mass of empty bulb	25.4 g
Mass of bulb filled with CO ₂ gas at temperature 17 °C and pressure 1.1 atm	26.5 g
Mass of bulb filled with water	619.5g

Calculate the relative molecular mass of CO₂ gas.

(40.06)

There is a difference between the value of the relative molecular mass calculated above and the value calculated from the molecular formula. Explain.

PSPM 2003 / 2004 (CLO3, C4)

16. Nitrogen gas is a real gas that deviates from ideality. It is the most abundant gas found in the atmosphere.

- a) Why does nitrogen gas behave as a real gas at room temperature?
b) State the conditions under which nitrogen gas can be liquefied.
c) How can nitrogen gas be made to behave as an ideal gas?

PSPM 2016 / 2017 (CLO3, C4)

17. A 3.0 L flask at 298 K contains a mixture of 2.0 mol helium gas and 3.0 mol xenon gas. The van der Waals constant for helium and xenon are given the following table:

Gas	a (L ² atm mol ⁻²)	b (L mol ⁻¹)
Helium	0.03421	0.02370
Xenon	4.194	0.05105

Given the van der Waals equation: $\left(P + \frac{n^2 a}{V^2} \right) (V - nb) = nRT$

- a) Which gas is expected to behave ideally and which gas exhibits a marked deviation from ideal behaviour?
b) Calculate the total pressure of the gas in the mixture using an ideal gas equation.

(40.76 atm)

- c) Using the van der Waals equation, the total pressure of the system is calculated to be 38.1 atm. Explain the reason behind the differences between this value and the pressure calculated in 17 (b).
PSPM 2017 / 2018 (CLO3, C4)

HOUR : 4

18. Describe properties of liquid below in terms of intermolecular forces, molecular arrangement, and molecular motion:
a) shape
b) volume
c) surface tension
d) viscosity
(CLO1, C2)
19. Explain the following processes based on kinetic molecular theory and intermolecular forces.
a) vaporisation
b) condensation
(CLO1, C2)
20. Define the term
a) vapour pressure
b) boiling point.
(CLO1, C1)
21. *"The boiling point of a liquid increases with the increase in the strength of intermolecular forces".*
Explain the statement.
PSPM JUNE 1999 (CLO3, C4)
22. The following table shows the boiling points of several liquids.

Liquid	Ethanal CH ₃ CHO	Ethanol CH ₃ CH ₂ OH	Methanol CH ₃ OH
Boiling Point (°C)	20	78	65

Arrange the liquids in the increasing order of vapour pressure. Explain.

PSPM 2001 / 2002 (CLO3, C4)

23. By applying kinetic concept, explain the process of:
a) freezing
b) melting
c) sublimation
d) deposition.
(CLO1, C2)
24. Differentiate between a crystalline solid and amorphous solid in terms of their particle arrangement and give example.
PSPM 2013 / 2014 (CLO1, C2)
25. Give an example for each of the following crystalline solids:
a) metallic
b) ionic
c) molecular covalent
d) giant covalent
(CLO1, C1)
26. Explain the following observation:

Carbon dioxide, CO₂ and diamond both form covalent bond. However, solid CO₂ is softer and exhibits lower melting point as compared to diamond.
PSPM 2015 / 2016 (CLO3, C3)